PQ-Auth Sentinel — Research Testing Report

Dataset: synthetic | Target: attack_type

Selected model and held-out test results

Measure	Result
Selected model	Soft Voting (RF + Extra Trees)
Test accuracy	98.98%
Test macro-F1	0.990
Test weighted F1	0.990
Train / validation / test rows	12000 / 4000 / 4000
Input feature count	7

Algorithm comparison

Algorithm	Validation Accuracy	Validation Macro F1	Test Accuracy	Test Macro F1	Status
Logistic Regression	0.992	0.991	0.987	0.987	OK
Decision Tree	0.986	0.986	0.987	0.987	OK
Random Forest	0.992	0.992	0.990	0.989	OK
Extra Trees	0.993	0.992	0.990	0.990	OK
Soft Voting (RF + Extra Trees)	0.993	0.993	0.990	0.990	OK

Research limitation

UNSW-NB15 is a network intrusion detection benchmark. Its metrics do not directly validate ML-DSA signatures or the application's seven authentication telemetry features. Report results on each dataset separately.